A novel method for measuring lung metabolism of precision-cut lung slices

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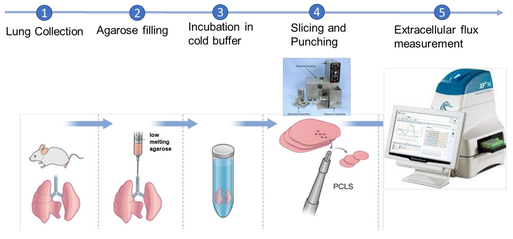
ABSTRACT

Pulmonary diseases are associated with changes in metabolism. Cellular metabolism is dynamically regulated by signaling cascades at molecular levels and associated with the pathology at organ level. Improved understanding of cellular metabolism is necessary to investigate its role in disease and identify the potential targets for therapy. We have developed a real-time method to adapt to Agilent XFe96 metabolic analyzer for precision-cut lung slices (PCLS) so that compound effects on metabolism can be screened across multiple technical replicates. PCLS were cut in 3 mm maximum diameter to allow for fitting and adhering to XFe96 plate wells, which were pre-coated with matrix gel. Once the PCLS were attached to the well bottom, the XFe96 cartridge probe can hold the tissue to measure, since the measurement volume is about 2-6 µl, 200 micrometers (µm) above the lung tissue. This method generated reproducible oxygen consumption and extracellular acidification data in precision-cut lung slices. We have optimized the conditions of oligomycin (OM) and Trifluoromethoxy carbonylcyanide phenylhydrazone (FCCP), rotenone and antimycin A to produce precision measurements. We have been able to utilize this technique to establish baseline metabolic rates for extracellular acidification and oxygen consumption and the dependency of those rates on mitochondrial function. In comparison with current cellular metabolic measurements, this method measures the whole tissue instead of an individual cell population. We have utilized this method to measure the effects of LPS treatment on PCLS metabolic function. Specifically, we have been able to demonstrate that LPS treatment not only leads to increased nitric oxide production in PCLS, but also reduces both glycolysis and spare respiratory capacity (SRC). This new method allows for the investigation of lung metabolism *ex vivo* and the screening of therapeutic targets for their interaction with lung metabolism.

INTRODUCTION

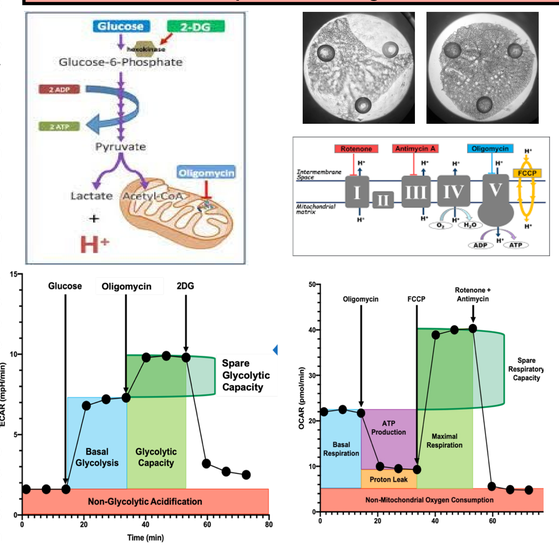
- Cells catabolize major nutrient substrates such as sugars, amino acids and fatty acids (FA) to introduce acetyl-CoA into the tricarboxylic acid (TCA) cycle to produce metabolites for the synthesis of nucleotides, proteins and lipids, and most importantly to generate energy.
- Cellular energetics is important for the maintenance of lung cell function under various pathophysiological conditions. Metabolic homeostasis is vital in regulating cell proliferation and differentiation, and studies have shown that metabolic dysregulation contributes to various inflammatory responses and apoptosis. These cellular processes have been implicated in various lung disease pathogenesis.
- understanding the metabolic processes underlying lung metabolite homeostasis in health and disease could identify new therapeutic targets for the treatment of lung infections.
- For more mechanistic understanding and a system with capabilities to investigate multiple regions of the lung, Precision Cut Lung Slices (PCLS) are one potential solution. Slices can be cultured from explanted human lung, diseased human lung and animal models of disease.
- PCLS contain all cell types found in lung of interest as well as accurately reflecting any changes to the underlying extracellular matrix associated with the disease.
- Most previous work has measured cellular metabolism of individual cell components in the lung. We attempt to measure cellular metabolism in PCLS, which can represent the whole lung metabolism in respiratory system.

METHODS



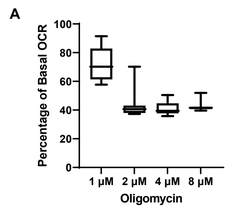
RESULTS

Experimental Design

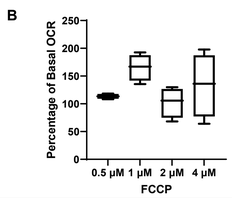


Optimization of mitochondrial inhibitors

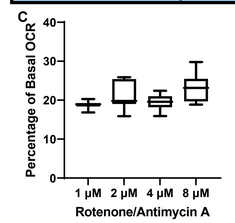
Oligomycin



FCCP



Rotenone/Antimycin A



Relations of Protein levels and Spare respiratory Capacity (SCR)

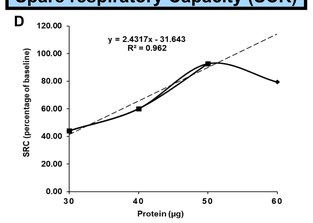


Figure 1. Optimization of respiration inhibitor concentrations. Box and whisker plots (generated using 3 mm diameter lung tissue punches) demonstrate the maximum response to increasing concentration of (A) the ATP synthase inhibitor oligomycin (B) the uncoupler FCCP and (C) the complex I inhibitor Rotenone and the complex III inhibitor Antimycin A. Loading size of the punches (D) is associated with the response by spare respiratory capacity (SRC).

Bioenergetic profiles in LPS challenged PCLS model

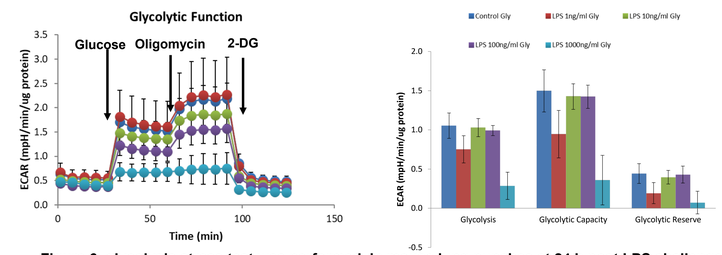


Figure 6. glycolysis stress test was performed in mouse lung punches at 24 h post LPS challenge.

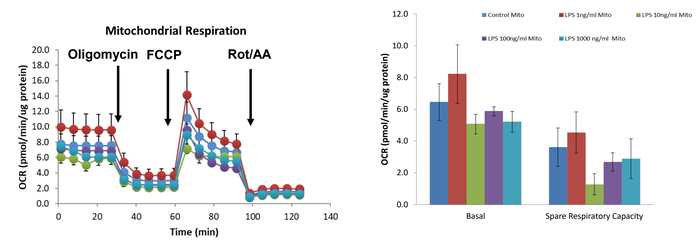


Figure 7. Mitochondrial function was measured by mito-stress test at 24 h post LPS challenge.

Glycolysis and Oxidative Phosphorylation (Mitochondria ETC) Tests

Glycolysis stress test

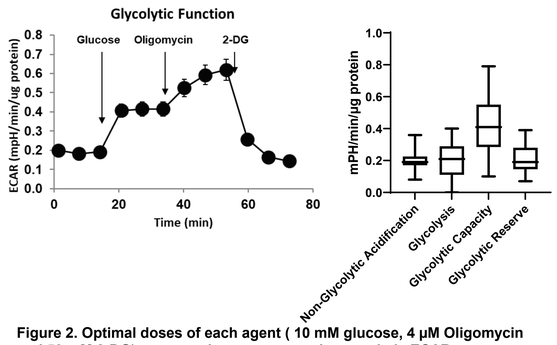


Figure 2. Optimal doses of each agent (10 mM glucose, 4 µM Oligomycin and 50 mM 2-DG) were used to generate a characteristic ECAR curve (glycolysis stress test) in mouse lung punches.

Mitochondrial stress test

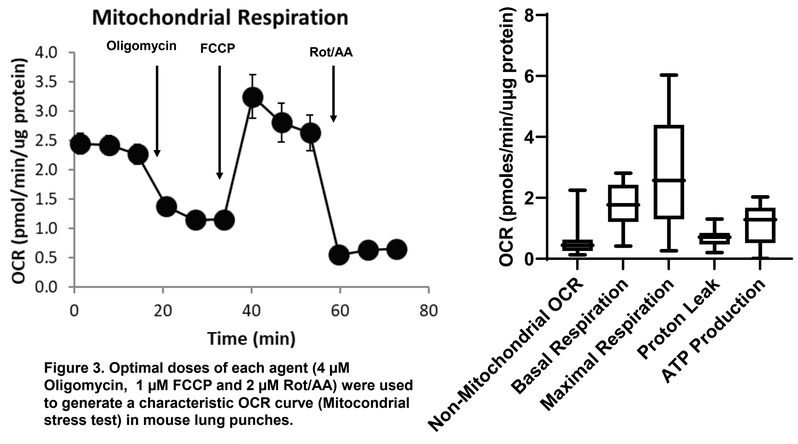


Figure 3. Optimal doses of each agent (4 µM Oligomycin, 1 µM FCCP and 2 µM Rot/AA) were used to generate a characteristic OCR curve (Mitocondrial stress test) in mouse lung punches.

Real-time ATP generation assay

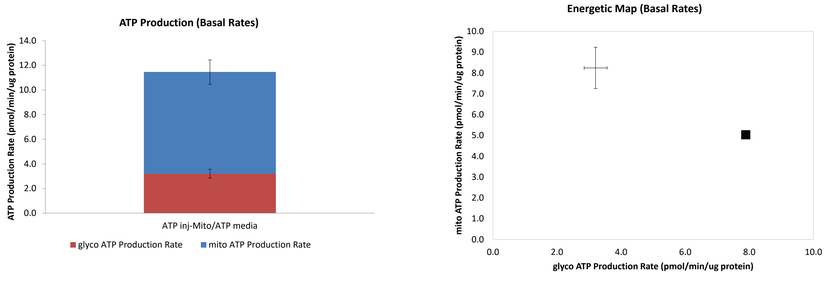


Figure 4. ATP production rate from OXPHOS and glycolysis. Real-time production of ATP was measured by injections of oligomycin and Rotenone/antimycin A in mouse lung punches.

LPS challenged PCLS model

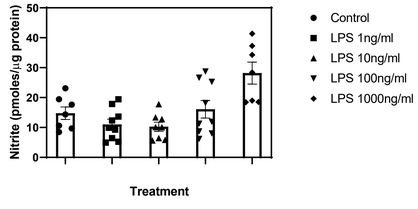


Figure 5. Nitric oxide production in LPS challenged PCLS. PCLSs were incubated with different concentrations of LPS for 24 h. Nitrite levels in the media were determined by Griess Assay.

NOTES

1. This protocol has been used to measure LPS and nitrogen mustard challenged PCLS. It was used to measure lung metabolism in different genetic background mice. It is also used to measure human lung metabolism by PCLS.
2. It is critical to adhere the PCLS cuts in Seahorse culture plate wells, before loading the culture plate, check one more time to make sure the tissue is not floating in the well.
3. The measurement is associated with the tissue size, if it is too small, the data would be not reproducible. Usually, one lung slice can obtain two or three punches.

SUMMARY

We have adopted a method to measure lung metabolism in an *ex vivo* system. The PCLS were cut to fit the size of XFe96 well culture plate and the OCR inhibitors were optimized and produced a stable reproducible measurement. Meanwhile, we produce a method which is easy to adapt to other solid organ systems, such as spleen, liver, heart, thymus, kidney, thyroid or brain. The disease is usually associated with the metabolic alterations in a system. Metabolic measurement in the tissue with drug treatment or transgenic animals will increase our understanding of disease and identification of therapeutic targets.

ACKNOWLEDGMENTS

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